A proof of a binomial-series conjecture of Zhi-Wei Sun

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Abstract

We prove a binomial-series conjecture of Zhi-Wei Sun whose value is $90/\pi$. A constant-term integral and an explicit exact differential reduce the double series to a period on a rational parameter curve. The Horn equations give a third-order differential equation identifying this period with the level-30 modular period of Lian and Yau. A Hecke correspondence for isogenies of degree 7 then determines the required complex multiplication value and its derivative. We give the full modular polynomial, exact finite certificates, and the convergence and branch arguments needed to justify the evaluation.

1 Introduction and notation

Sun proposed the following identity immediately after Conjecture 1.7 in [1, p. 11]. The purpose of this paper is to prove this conjecture.

Theorem 1.1 (Sun's binomial-series conjecture)**.** *The series below converges absolutely, even after the absolute values of all its summands indexed by (n,k) are taken, and*

$$\begin{aligned}\sum_{n=0}^{\infty}\frac{357n+103}{2160^n}\binom{2n}{n}a_n&=\frac{90}{\pi},\\ a_n&=\sum_{k=0}^{n}\binom{n}{k}\binom{n+2k}{2k}\binom{2k}{k}(-324)^{n-k}.\end{aligned}\tag{1.1}$$

Our proof has three reductions. First, a two-variable generating function converts the series into a period and a radial derivative. Second, an exact differential and the Horn system reduce both to a single modular period. Third, a finite reconstruction of a Hecke modular equation gives the special value. The only imported period identity is the level-30 identity of Lian–Yau [2]; its group, normalization, and square-root branch are checked below. In particular, the proof does not infer an identity from agreement of finitely many Taylor coefficients.

Throughout, m,n,k denote nonnegative integers, and $\binom{n}{k}=n!/(k!(n-k)!)$ for $0\le k\le n$. For a variable r, write $\theta_r=r\,\mathrm{d}/\mathrm{d}r$. For complex variables u,v, put $\Theta=u\partial_u+v\partial_v$. The notation $[x^j]$ means coefficient extraction and $\mathrm{CT}_{x,y}$ means the coefficient of x^0y^0 in the stated Laurent expansion. All parameter derivatives of F are taken with the other original variable fixed.

Square roots are specified by analytic continuation from the indicated initial branch. Matrices of positive determinant act on $\mathbb{H} = \{\tau \in \mathbb{C} : \operatorname{Im}\tau > 0\}$ by fractional linear transformations; scalar multiples give the same transformation. Composition of differential operators includes differentiation of their coefficients.

2 The generating function and absolute convergence

Define, initially near $(0,0)$,

$$F(u,v) = \sum_{m,k\geq 0} \binom{2m+2k}{m+k} \frac{(m+3k)!}{m!(k!)^3} u^m v^k, \qquad \Theta = u\partial_u + v\partial_v. \tag{2.1}$$

The elementary identity

$$\binom{n}{k}\binom{n+2k}{2k}\binom{2k}{k} = \frac{(n+2k)!}{(n-k)!(k!)^3}$$

and the substitution $m = n-k$ give the formal reduction

$$S = (357\Theta + 103)F\left(-\frac{3}{20}, \frac{1}{2160}\right). \tag{2.2}$$

where S denotes the left-hand side of (1.1). The next proposition justifies this reduction analytically.

Proposition 2.1 (Constant term and convergence)**.** *Put*

$$\mathcal{D} = \left\{(u,v) \in \mathbb{C}^2 : 4\left(|u| + \frac{10000}{99}|v|\right) < \frac{4}{5}\right\}.$$

On $\mathcal{D}$, F and every fixed polynomially weighted coefficient series converge absolutely and locally uniformly. Moreover,

$$F(u,v) = \mathrm{CT}_{x,y}\, \frac{1}{a}\left(1 - \frac{4(u+v/(xy))}{a}\right)^{-1/2}. \tag{2.3}$$

where $a = 1-x-y$ and the inner inverse square root is expanded with value 1 at $u = v = 0$. Formula (2.2) and all subsequent parameter differentiations at the target point are valid.

Proof. Set $a = 1-x-y$. The coefficient of $u^m v^k$ in the right-hand side of (2.3) is

$$\binom{2m+2k}{m+k}\binom{m+k}{k}[x^k y^k](1-x-y)^{-m-k-1}.$$

Since

$$[x^k y^k](1-x-y)^{-m-k-1} = \frac{(m+3k)!}{(m+k)!(k!)^2},$$

this is precisely the coefficient in (2.1).

Use the positively oriented product torus

$$\mathcal{T} = \{|x| = 9/100,\ |y| = 11/100\},$$

with the product orientation of the two counterclockwise circles. There $|x+y| \le 1/5$ and $|xy| = 99/10000$. The binomial expansion in (2.3) and the expansions in x, y are dominated by the corresponding positive series. At $(u, v) = (-3/20, 1/2160)$ their ratio bound is

$$\frac{4(|u| + |v|/|xy|)}{1 - |x| - |y|} = \frac{10519}{10692} < 1. \tag{2.4}$$

More generally the same bound is strictly less than 1 on $\mathcal{D}$, uniformly on compact subsets. To include a polynomial weight of degree d, choose a slightly larger pair of absolute parameter values still in $\mathcal{D}$; the exponential gain dominates $(1+m+k)^d$. This proves local normal convergence of the series and its derivatives. Absolute convergence permits the change from (n, k) to (m, k) and termwise application of $357\Theta + 103$.

Coefficient extraction on $\mathcal{T}$ now yields

$$F(u, v) = \frac{1}{(2\pi \mathrm{i})^2} \int_{\mathcal{T}} \frac{\mathrm{d}x \wedge \mathrm{d}y}{xy\sqrt{(1-x-y)(1-x-y-4u-4v/(xy))}}. \tag{2.5}$$

Here the denominator square root means

$$(1-x-y)\left(1 - \frac{4(u+v/(xy))}{1-x-y}\right)^{1/2},$$

with the binomial branch, so it is single-valued and nonzero near the torus for the parameters under consideration. □

We restrict the parameters to

$$U(r) = -\frac{r(2+3r)}{4(1+2r)}, \qquad V(r) = \frac{r^4}{16(1+2r)}, \qquad g(r) = F(U(r), V(r)). \tag{2.6}$$

For $0 \le r \le 1/3$, both $|U(r)|$ and $V(r)$ increase to $3/20$ and $1/2160$, respectively. The strict majorant therefore holds along the entire interval. The fixed-torus representation shows that g and every parameter derivative needed below are holomorphic on a complex neighborhood of this interval.

3 Radial and tangential derivatives

Lemma 3.1 (An exact differential)**.** *Along* (2.6)*,*

$$Cg + \alpha F_u(U, V) + \beta F_v(U, V) = 0, \tag{3.1}$$

where

$$\begin{aligned} C &= 3r(1+2r)(r+2), \\ \alpha &= -\frac{3r(3r+2)(r^2+6r+2)}{4}, \\ \beta &= \frac{3r^4(3r^2+4r-2)}{16}. \end{aligned} \tag{3.2}$$

The identity holds on a complex neighborhood of $[0, 1/3]$*.*

Proof. We first take $0 < r \le 1/3$. In (2.5), set

$$s = x + y, \quad p = xy, \quad c = 1 - 4U, \quad b = 4V, \qquad \mathcal{W} = p\sqrt{(1-s)(c-s-b/p)},$$

and then put

$$X = (c-s)p, \qquad Y = (c-s)(x-y)\mathcal{W}.$$

Calculation in the original coordinates gives

$$\begin{aligned} Y^2 = P(s,X) &:= (1-s)X\{s^2(c-s) - 4X\}(X-b), \\ \frac{\mathrm{d}s \wedge \mathrm{d}X}{Y} &= \frac{\mathrm{d}x \wedge \mathrm{d}y}{\mathcal{W}}. \end{aligned} \tag{3.3}$$

The second identity is an identity of pulled-back differential forms. All integration is still over the original torus, so a degree of the algebraic coordinate map does not occur.

Here is a rational certificate for the required exact form. Define

$$\begin{aligned} q_s &= -\frac{r^2}{4(r+1)^2}, & \sigma &= -\frac{r^2}{1+2r}, \\ X_0 &= q_s s(s-c)(s-1), & h &= s(s-c)(s-1)(s-\sigma)(s-r-1), \\ K &= \frac{3r^2(2r+1)^2}{2(r+1)^3}, & R_0 &= -\frac{3(2r+1)(s-1)(r^3+r^2+4rs+2s)}{r+1}, \end{aligned}$$

and

$$\begin{aligned} T_1 &= -\frac{3(2r+1)(3r^3+7r^2+12rs-4r+6s-4)}{r+1}, \\ T_0 &= \frac{3r^2(r-s+1)^2(r^2+2rs+s)(6rs-3r+3s-1)}{2(r+1)^3}, \\ R &= \frac{Kh}{X-X_0} + R_0, \\ \mathcal{A} &= \frac{K(\partial_s X_0)h}{X-X_0} + T_1 X + T_0 - \frac{4\alpha X}{c-s}. \end{aligned}$$

With r, s, X treated as independent variables, these functions satisfy

$$R_s + \mathcal{A}_X - \frac{RP_s + \mathcal{A}P_X}{2P} = C + \frac{2\alpha X}{(c-s)(X-b)} + \frac{2\beta}{X-b}. \tag{3.4}$$

This is a finite rational identity: substitute the displayed formulas, clear denominators, and compare polynomial coefficients. The ancillary code also checks it exactly. The term $-4\alpha X/(c-s)$ is needed because at fixed x, y one has $X_u = -4X/(c-s)$. Explicitly, the coordinate correction is

$$\frac{2s^2}{s^2(c-s)-4X} - \frac{4}{c-s} + \frac{2XP_X}{(c-s)P} = \frac{2X}{(c-s)(X-b)}. \tag{3.5}$$

Define the one-form

$$\Omega = \frac{R\,\mathrm{d}X - \mathcal{A}\,\mathrm{d}s}{Y}.$$

Its exterior derivative is the right-hand side of (3.4) times $\mathrm{d}s \wedge \mathrm{d}X/Y$. We check that its pullback has no poles near $\mathcal{T}$. The convergence bound makes $\mathcal{W}$ nonzero, and the unequal circle radii

make $x-y$ nonzero. Also $1-s$, $c-s$ and $X-b$ do not vanish: here $c\ge 1$, $|s|\le 1/5$, and the second factor of $\mathcal{W}^2$ is nonzero. For the remaining denominator, $|q_s|\le 1/64$ and hence

$$\left|\frac{X_0}{c-s}\right| = |q_s s(s-1)| \le \frac{3}{800} < \frac{99}{10000} = |p|.$$

Thus $X - X_0 \neq 0$. All inequalities are strict and persist in suitable complex neighborhoods; the pullback of Ω is single-valued there. Stokes' theorem on the closed torus gives $\int_{\mathcal{T}} \mathrm{d}\Omega = 0$.

At fixed x, y, the multipliers obtained by differentiating the two-form for F with respect to u, v are, respectively,

$$\frac{2X}{(c-s)(X-b)}, \qquad \frac{2}{X-b}.$$

Integrating (3.4) therefore proves (3.1). All terms are holomorphic near the interval, including at $r=0$, so the identity extends there by analytic continuation. □

Corollary 3.2 (Reduction at the target point)**.** *At $r = 1/3$,*

$$\Theta F = \frac{15\theta_r g - 7g}{51}, \qquad S = 3(35\theta_r + 18)g(1/3). \tag{3.6}$$

In the first equality, F is evaluated at $(U(1/3), V(1/3))$.

Proof. The radial relation becomes

$$\frac{35}{9}g - \frac{37}{12}F_u - \frac{1}{1296}F_v = 0.$$

The chain rule and the definition of Θ give

$$\theta_r g = -\frac{7}{50}F_u + \frac{1}{600}F_v, \qquad \Theta F = -\frac{3}{20}F_u + \frac{1}{2160}F_v.$$

Eliminating F_u, F_v proves the first equality; multiplying it by 357 and adding $103g$ proves the second. □

4 The differential equation of the restricted period

Write $A = u\partial_u$ and $B = v\partial_v$. The adjacent factorial ratios of the coefficients of (2.1) give the Horn equations

$$\begin{aligned} &H_1 F = 0, \quad H_1 = A(A+B) - 2u(2A+2B+1)(A+3B+1),\\ &H_2 F = 0, \quad H_2 = B^3(A+B)\\ &\qquad\qquad - 2v(2A+2B+1)\prod_{j=1}^{3}(A+3B+j). \end{aligned} \tag{4.1}$$

For example, if $c_{m,k}$ denotes that coefficient, then

$$\frac{c_{m,k}}{c_{m-1,k}} = \frac{2(2m+2k-1)(m+3k)}{m(m+k)} \quad (m\ge 1),$$

and

$$\frac{c_{m,k}}{c_{m,k-1}} = \frac{2(2m+2k-1)(m+3k)(m+3k-1)(m+3k-2)}{k^3(m+k)} \quad (k\ge 1).$$

These are precisely the two coefficient recurrences. At $m=0$ or $k=0$ the corresponding left Euler factor vanishes, so the equations include the coordinate boundaries.

Proposition 4.1 (Restriction of the Horn system)**.** *The function g satisfies $L_3 g = 0$, where*

$$L_3 = \sum_{j=0}^{3} p_j(r)\theta_r^j, \tag{4.2}$$

and

$$\begin{aligned} p_0 &= \frac{r(144r^5 + 159r^4 + 9r^3 - 45r^2 - 16r - 2)}{48}, \\ p_1 &= \frac{r(2r+1)(312r^4 + 151r^3 - 150r^2 - 106r - 18)}{96}, \\ p_2 &= \frac{r(2r+1)^3(18r^2 - 13r - 10)}{32}, \\ p_3 &= \frac{(r-2)(2r+1)^3(3r+2)(4r+1)}{96}. \end{aligned} \tag{4.3}$$

Proof. We give an exact elimination, including the removal of an extraneous first-order factor.

Step 1: normal coordinates. Put

$$u = U(r), \qquad v = V(r) + t, \qquad a = \frac{1}{U'}, \qquad d = -\frac{V'}{U'}.$$

Thus $\partial_u = a\partial_r + d\partial_t$ and $\partial_v = \partial_t$. These coordinates are valid for small nonzero r, since

$$U' = -\frac{3r^2 + 3r + 1}{2(2r+1)^2} \neq 0.$$

In ordinary partial derivatives the coefficients of H_1 are

$$\begin{aligned} &c_{20} = (1-4u)u^2, \quad c_{11} = (1-16u)uv, \quad c_{02} = -12uv^2, \\ &c_{10} = u(1-10u), \quad c_{01} = -22uv, \qquad\quad c_{00} = -2u. \end{aligned}$$

After the coordinate change they are

$$\begin{aligned} C_{20} &= c_{20}a^2, & C_{11} &= 2c_{20}ad + c_{11}a, \\ C_{02} &= c_{20}d^2 + c_{11}d + c_{02}, & C_{10} &= c_{20}aa' + c_{10}a, \\ C_{01} &= c_{20}ad' + c_{10}d + c_{01}, & C_{00} &= c_{00}, \end{aligned}$$

Here every c_{ij} is evaluated at $(U(r), V(r)+t)$, and primes denote differentiation with respect to r. In particular,

$$C_{02}(r,0) = \frac{3r^8(r+1)(3r+2)(9r^2+8r+2)}{256(2r+1)^2(3r^2+3r+1)^2} \neq 0 \quad (0 < r \leq 1/3).$$

Step 2: eliminating normal derivatives. For $(i,j) \neq (0,2)$ set $Q_{ij} = -C_{ij}/C_{02}$, and define the ordinary differential operators J_j by

$$J_0 g = g, \qquad J_1 g = -\frac{Cg + \alpha a g'}{\alpha d + \beta}, \tag{4.4}$$

$$J_{n+2} g = \sum_{(i,j)\neq(0,2)} \sum_{k=0}^{n} \binom{n}{k} (\partial_t^k Q_{ij})(r,0) \left(\frac{\mathrm{d}}{\mathrm{d}r}\right)^i J_{n-k+j} g, \qquad n = 0, 1, 2. \tag{4.5}$$

The outer sum runs over the six indices of C_{ij} except $(0,2)$. Here J_1 follows from Lemma 3.1, and its denominator is

$$\alpha d + \beta = -\frac{15r^4(r+1)(9r^2+8r+2)}{16(3r^2+3r+1)} \neq 0.$$

Let $\widetilde{F}(r,t) = F(U(r), V(r)+t)$. Solving $H_1 F = 0$ for $\partial_t^2 \widetilde{F}$ and differentiating n times in t gives (4.5) by the product rule. Induction shows that $J_j g = (\partial_t^j \widetilde{F})(r,0)$ for $0 \le j \le 4$.

Step 3: the exact operator factorization. Transform the factored operator H_2 to (r,t) coordinates, put all partial derivatives to the right, and replace $\partial_r^i \partial_t^j \widetilde{F}|_{t=0}$ by $(\mathrm{d}/\mathrm{d}r)^i J_j g$. Call the resulting operator N; then $Ng = 0$. With $D_0 = 9r^2 + 8r + 2$, exact rational arithmetic gives

$$N = \left(q_1 \frac{\mathrm{d}}{\mathrm{d}r} + q_0 \right) L_3, \tag{4.6}$$

where

$$\begin{aligned} q_1 &= -\frac{3r(2r+1)}{5D_0^2}, \\ q_0 &= \frac{r(1944r^5 + 5769r^4 + 6588r^3 + 3675r^2 + 1010r + 110)}{15(r+1)(2r+1)^2 D_0^3}. \end{aligned} \tag{4.7}$$

All coefficients stand to the left of derivatives; the product is operator composition. The preceding recursion specifies N completely, so (4.6) is verified by expanding and checking its five ordinary-derivative coefficients. This check is also included in the ancillary files.

Step 4: removing the first-order factor. Set $E = L_3 g$. For small nonzero r, the factorization gives $E' = \kappa E$, with

$$\kappa = -\frac{q_0}{q_1} = \frac{1944r^5 + 5769r^4 + 6588r^3 + 3675r^2 + 1010r + 110}{9(r+1)(2r+1)^3(9r^2+8r+2)}.$$

Both E and κ are holomorphic at zero. Since $p_0(0) = p_1(0) = p_2(0) = 0$, $p_3(0) = -1/24$, and $\theta_r^j g(0) = 0$ for $j \ge 1$, we have $E(0) = 0$. The first-order equation extends to zero; uniqueness of its holomorphic initial-value problem gives $E \equiv 0$. Finally, $L_3 g$ is holomorphic near the entire interval, so the identity theorem extends the result along $[0, 1/3]$. □

Define $w(z)$ as the Taylor solution with $w(0) = 1$ of

$$\begin{aligned} \mathcal{L} = {} & \theta_z^3 - z(1+2\theta_z)(4+7\theta_z+7\theta_z^2) \\ & + z^2(1+\theta_z)(72+106\theta_z+53\theta_z^2) \\ & - 14z^3(1+\theta_z)(2+\theta_z)(3+2\theta_z) \\ & - 24z^4(2+\theta_z)(3+2\theta_z)(5+2\theta_z), \qquad \mathcal{L}w = 0. \end{aligned} \tag{4.8}$$

The coefficient of its new Taylor coefficient at order $n \ge 1$ is n^3, which proves existence and uniqueness as a formal series; the regular singular Frobenius theorem gives convergence near zero.

Corollary 4.2 (Identification of the period)**.** *Along* $0 \le r \le 1/3$,

$$g(r) = \sqrt{1+2r}\, w(-r/2). \tag{4.9}$$

The square root is positive for real $r \geq 0$, and $w(-1/6)$ means analytic continuation along the negative real segment.

Proof. Direct differentiation gives the full operator identity

$$L_3\{\sqrt{1+2r}\, w(-r/2)\} = -\frac{(1+2r)^{5/2}}{24}(\mathcal{L}w)(-r/2). \tag{4.10}$$

In $L_3 g = 0$ the multiplier of the new Taylor coefficient at order $n \geq 1$ is $-n^3/24$. Thus its holomorphic solution is determined by the constant coefficient. Both sides of (4.9) have value 1 at $r = 0$, so they agree near zero. The coefficient of the highest ordinary derivative is $r^3 p_3(r)$, which does not vanish for $0 < r \leq 1/3$; continuation along this interval is therefore valid. No convergence of the initial Taylor series at $z = -1/6$ is being assumed. □

5 The modular parameter and its normalization

For $\tau \in \mathbb{H}$, put $q = e^{2\pi i\tau}$ and define

$$\eta(\tau) = e^{\pi i\tau/12} \prod_{n\geq 1}(1-q^n).$$

The exponential fixes the fractional power $q^{1/24}$. We use

$$\begin{gathered} u_*(\tau) = \left[\frac{\eta(\tau)\eta(6\tau)\eta(10\tau)\eta(15\tau)}{\eta(2\tau)\eta(3\tau)\eta(5\tau)\eta(30\tau)}\right]^3, \\ T = u_* + 6 + u_*^{-1}, \qquad z = T^{-1}, \qquad D = \frac{1}{2\pi i}\frac{\mathrm{d}}{\mathrm{d}\tau} = q\frac{\mathrm{d}}{\mathrm{d}q}, \\ P_0(z) = (1-8z)(1-4z)(1+z)(1-3z). \end{gathered} \tag{5.1}$$

Let Γ be the subgroup of $\mathrm{PSL}_2(\mathbb{R})$ generated by $\Gamma_0(30)$ and the determinant-normalized matrices

$$W_2 = \begin{pmatrix} 8 & 1 \\ 30 & 4 \end{pmatrix}, \quad W_3 = \begin{pmatrix} 3 & 2 \\ 30 & 21 \end{pmatrix}, \quad W_5 = \begin{pmatrix} 25 & 4 \\ 30 & 5 \end{pmatrix}. \tag{5.2}$$

Thus $\Gamma = \Gamma_0(30)^+$, with all Atkin–Lehner involutions adjoined. We use the term Hauptmodul for a meromorphic coordinate on its compactified quotient, equivalently a generator of its meromorphic function field with degree one.

Proposition 5.1 (The eta-quotient Hauptmodul)**.** *The function T is a Hauptmodul for Γ, has no pole in $\mathbb{H}$, and has just one simple pole on the compactified quotient. Its expansion at infinity is*

$$u_* = q^{-1} \prod_{\gcd(n,30)=1} (1-q^n)^3, \qquad T = q^{-1} + 3 + 4q + 2q^2 + O(q^3). \tag{5.3}$$

Proof. For $\delta = 1, 2, 3, 5, 6, 10, 15, 30$, the eta exponents of u_* are

$$(r_\delta) = (3, -3, -3, -3, 3, 3, 3, -3).$$

They satisfy

$$\sum r_\delta = 0, \quad \sum \delta r_\delta = -24, \quad \sum (30/\delta) r_\delta = 24, \quad \prod \delta^{r_\delta} = 1.$$

The eta-quotient criterion [3, Theorems 1.64–1.65], also stated in [4, Theorems 1.1–1.2], makes u_* a weight-zero modular function on $\Gamma_0(30)$ with trivial character. Only modularity and meromorphy at the cusps are needed; negative exponents are allowed. Since 30 is squarefree, its order at a cusp with denominator $d \mid 30$ is

$$\frac{30}{24d} \sum_{\delta \mid 30} \frac{\gcd(d,\delta)^2}{\delta} r_\delta.$$

The eight orders, in the stated denominator order, are $(1,-1,-1,-1,1,1,1,-1)$.

The eta transformation formula [5, §23.18] gives

$$u_*(W_e\tau) = u_*(\tau)^{-1}, \qquad e = 2,3,5. \tag{5.4}$$

For completeness, here is a finite specification of the multiplier check. Write $W_e = (ea, b; 30c, ed)$ and, for each $\delta \mid 30$, set $h_\delta = \gcd(\delta, e)$, $m = \delta e / h_\delta^2$. Then

$$\delta W_e \tau = \begin{pmatrix} h_\delta a & (\delta/h_\delta) b \\ 30 h_\delta c/(e\delta) & ed/h_\delta \end{pmatrix} (m\tau).$$

This integer matrix has determinant one. For an integer matrix $(A, B; C, D)$ with $C > 0$, the required formula is

$$\eta\left(\frac{A\tau + B}{C\tau + D}\right) = e^{\pi \mathrm{i}((A+D)/(12C) - s(D,C) - 1/4)} (C\tau + D)^{1/2} \eta(\tau),$$

with the principal square root and the Dedekind sum

$$s(D,C) = \sum_{j=1}^{C-1} \left(\frac{j}{C} - \frac{1}{2}\right) \left(\left\{\frac{Dj}{C}\right\} - \frac{1}{2}\right).$$

Here $\{x\} = x - \lfloor x \rfloor$; no term Dj/C is integral in this sum because $\gcd(C, D) = 1$. Weighted by r_δ, the multiplier exponents for $e = 2, 3, 5$ all sum to 0, and $\prod h_\delta^{r_\delta} = 1$. The permutation $\delta \mapsto m$ negates the eta exponents, proving (5.4).

Consequently T is Γ-invariant. The standard squarefree Atkin–Lehner cosets are indexed by $e \mid 30$, and their actions toggle the prime divisors of the cusp denominator. Thus the eight cusps of $\Gamma_0(30)$ form one orbit. Eta has no zero in $\mathbb{H}$, so T has no interior pole. At each old cusp exactly one of u_* and u_*^{-1} has a simple pole, with no cancellation. The descended pole also has order one: its pullback order is its order times the positive ramification index. A meromorphic function on a compact Riemann surface with a unique simple pole has degree one onto $\mathbb{P}^1$. This proves the Hauptmodul assertion. Finally, collecting the eta factors gives the product and the expansion in (5.3). □

Theorem 5.2 (Lian–Yau period identity, with the present normalization)**.** *For* $\operatorname{Im}\tau$ *sufficiently large,*

$$w(z(\tau)) = \frac{Dz}{z\sqrt{P_0(z)}}. \tag{5.5}$$

The branch of $\sqrt{P_0(z)}$ *has value* 1 *at* $z = 0$*.*

Proof. The level-30 row of [2, printed p. 9] has exactly the operator (4.8). Equations (4.6)–(4.7) and Theorem 4.2, equation (4.14), give the squared period identity for its mirror map. The sign is fixed by $w(0) = 1$. We check that this mirror map is our $z = 1/T$.

The Frobenius normalization in [2, (4.12)–(4.13)] is $q = z\exp(h_0(z)/\widetilde{w}(z))$, with $h_0(0) = 0$ and $\widetilde{w}(0) = 1$. It gives $z = q + O(q^2)$, fixing the otherwise undetermined multiplicative constant of q. Write $z = q + bq^2 + O(q^3)$. From (4.8), $w(z) = 1 + 4z + O(z^2)$, and $P_0(z) = 1 - 14z + O(z^2)$. Hence

$$\frac{Dz}{z\sqrt{P_0(z)}} = 1 + (b+7)q + O(q^2),$$

so $b = -3$ and the reciprocal mirror map is $q^{-1} + 3 + O(q)$. It is the Hauptmodul for the same group in the Lian–Yau row, with its pole at the same cusp. Two such coordinates with the same leading and constant terms differ by a holomorphic function on the compact quotient vanishing at the cusp, and therefore are equal. Proposition 5.1 identifies it with T. □

6 A Hecke modular equation from finite principal parts

Use the eight determinant-7 matrices

$$H_0 = \begin{pmatrix} 7 & 0 \\ 0 & 1 \end{pmatrix}, \qquad H_{j+1} = \begin{pmatrix} 1 & j \\ 0 & 7 \end{pmatrix}, \quad 0 \le j \le 6. \tag{6.1}$$

Proposition 6.1 (The Hecke modular polynomial)**.** *There is a polynomial* $\Phi(X, Y) \in \mathbb{Z}[X, Y]$*, monic of degree eight in* Y*, characterized by*

$$\Phi(T(\tau), Y) = \prod_{i=0}^{7} \{Y - T(H_i\tau)\} \tag{6.2}$$

It is symmetric, has degree eight in each variable, and is given completely in Appendix A. It satisfies

$$\begin{aligned} \Phi(X, X) = & -X^2(X-24)(X-6)(X-4)^2(X+2)^2(X+6) \\ & \cdot (X^2 - 4X - 4)(X^3 - 10X^2 + 28X + 4), \end{aligned} \tag{6.3}$$

and

$$\Phi_X(-6, -6) = 429649920000, \qquad \frac{\Phi_{XX} - \Phi_{XY}}{\Phi_X}(-6, -6) = \frac{61}{315}. \tag{6.4}$$

Proof. Since $\gcd(7, 30) = 1$, the usual Hecke coset decomposition makes $\Gamma_0(30)$ permute the eight functions $T(H_i\tau)$. The same holds for the three Atkin–Lehner generators. Explicitly, in

order $i = 0, \dots, 7$, their permutations are

$$\begin{aligned} \sigma_2 &= (3,2,5,7,0,6,1,4), \\ \sigma_3 &= (1,4,7,0,2,5,6,3), \\ \sigma_5 &= (7,2,6,1,5,3,0,4). \end{aligned} \tag{6.5}$$

These lists give $\sigma_e(i)$, rather than cycle notation. For each e and i, the matrix $H_i W_e H_{\sigma_e(i)}^{-1} W_e^{-1}$ is integral, has determinant one, and has lower-left entry divisible by 30. This finite matrix check proves the asserted permutation property for the full group.

It follows that every coefficient of the product in (6.2) is a Γ-modular function without an interior pole, hence a polynomial in T by Proposition 5.1. At the unique cusp the eight roots have Puiseux pole orders 7 and seven copies of $1/7$. Thus the kth elementary symmetric function has pole order at most $7 + (k-1)/7$. Its order is integral, so its degree in T is at most seven for $1 \le k \le 7$ and at most eight for $k = 8$.

We now reconstruct the actual coefficients, rather than fit an unproved polynomial relation. Let S_m be the mth power sum of the eight roots. It is a polynomial in T of degree at most $7m$. By the seventh-root-of-unity filter, its nonpositive Laurent part is that of

$$T(q^7)^m + 7 \sum_n [q^{7n}] T(q)^m \, q^n. \tag{6.6}$$

The sum is over integers n for which the indicated coefficient is defined. A polynomial in $T = q^{-1} + 3 + \cdots$ of degree at most $7m$ is uniquely determined by its nonpositive Laurent part: subtract the coefficient of q^{-d} times T^d, successively for $d = 7m, 7m-1, \dots, 0$. Any remaining difference is holomorphic on the compact quotient and vanishes at the cusp.

For $1 \le m \le 8$ this gives S_m exactly. Newton's identities then give the elementary symmetric polynomials E_k:

$$E_0 = 1, \qquad k E_k = \sum_{j=1}^{k} (-1)^{j-1} E_{k-j} S_j \quad (1 \le k \le 8), \qquad \Phi = \sum_{k=0}^{8} (-1)^k E_k(X) Y^{8-k}. \tag{6.7}$$

The largest degree needed is 56; the product for qT truncated through q^{64} is therefore more than sufficient. Indeed, computing the nonpositive part of $T^d = q^{-d}(qT)^d$ only requires qT through degree $d \le 56$. All divisions by k in this reconstruction are exactly divisible in the integers for the displayed data. The resulting array is Appendix A; its transpose equals itself, proving symmetry. Substituting $Y = X$, factoring, and differentiating that array gives (6.3) and (6.4). This supplies both the finite computation and the bound making it sufficient. Here 7 is the degree of the underlying isogenies and the determinant of H_i; the polynomial degree in either variable is 8. □

7 The complex multiplication value and its derivative

Put

$$\tau_0 = \frac{-15 + \mathrm{i}\sqrt{105}}{30}, \qquad \gamma = \begin{pmatrix} -15 & -11 \\ 30 & 15 \end{pmatrix}. \tag{7.1}$$

Then $\det\gamma = 105$ and $\gamma\tau_0 = \tau_0$. We have

$$\begin{aligned} \gamma &= W_{15}M_7, \qquad W_{15} = \begin{pmatrix} 15 & 7 \\ 30 & 15 \end{pmatrix}, \\ M_7 &= \begin{pmatrix} -29 & -18 \\ 60 & 37 \end{pmatrix} = \begin{pmatrix} -29 & 14 \\ 60 & -29 \end{pmatrix} H_5. \end{aligned} \tag{7.2}$$

Here W_{15} is an Atkin–Lehner matrix and the last determinant-one matrix lies in $\Gamma_0(30)$. Thus $T(\gamma\tau) = T(H_5\tau)$, and

$$\Phi(T(\tau), T(\gamma\tau)) = 0. \tag{7.3}$$

Lemma 7.1 (Selection of the CM root)**.** *At τ_0, one has $T(\tau_0) = -6$ and $z(\tau_0) = -1/6$. Moreover, $T'(\tau_0) \neq 0$, where the prime denotes $\mathrm{d}/\mathrm{d}\tau$.*

Proof. At the fixed point, (7.3) implies $\Phi(T(\tau_0), T(\tau_0)) = 0$. We select its root by rational bounds, rather than by a numerical recognition argument. Write

$$q(\tau_0) = -\rho, \qquad \rho = e^{-\pi\sqrt{105}/15}.$$

The elementary bounds $3.14 < \pi < 22/7$ and $10.24 < \sqrt{105} < 10.25$, followed by the degree-31 and degree-32 Taylor bounds for e^{-x}, give

$$\frac{1167}{10000} < \rho < \frac{1173}{10000}. \tag{7.4}$$

More explicitly, set

$$a = \frac{(314/100)(1024/100)}{15}, \qquad b = \frac{(22/7)(1025/100)}{15}.$$

Taylor's formula with signed remainder gives the bounds from the finite rational inequalities

$$\frac{1167}{10000} < \sum_{j=0}^{31} \frac{(-b)^j}{j!}, \qquad \sum_{j=0}^{32} \frac{(-a)^j}{j!} < \frac{1173}{10000}.$$

Every positive integer coprime to 30 is odd, so $u_* < 0$ and

$$|u_*| = \rho^{-1}(1+\rho)^3 \prod_{n\geq 7,\,(n,30)=1} (1+\rho^n)^3.$$

Put $\rho_{\mathrm{hi}} = 1173/10000$, $\rho_{\mathrm{lo}} = 1167/10000$, $E_0 = 3\rho_{\mathrm{hi}}^7/(1-\rho_{\mathrm{hi}})$, and $f(t) = (1+t)^3/t$. The tail product is between 1 and $(1-E_0)^{-1}$, since $\prod(1+a_j) \leq e^{\sum a_j} \leq (1-\sum a_j)^{-1}$ for $a_j \geq 0$ with $\sum a_j < 1$. Since f decreases on this interval,

$$f(\rho_{\mathrm{hi}}) < |u_*| < \frac{f(\rho_{\mathrm{lo}})}{1-E_0}.$$

These rational bounds give

$$-\frac{61}{10} < T(\tau_0) < -\frac{59}{10}. \tag{7.5}$$

In this interval all factors in (6.3) except $X+6$ are nonzero. The quadratic roots $2 \pm 2\sqrt{2}$ are greater than -1, and $X^3 - 10X^2 + 28X + 4 < 0$ there, since $X < -5$. Therefore $T(\tau_0) = -6$.

We next justify the nonzero derivative. Each element of Γ has an integral representative of determinant $e \mid 30$ with its lower-left entry divisible by 30, by the squarefree Atkin–Lehner coset description. If $(a, b; c, d)$ fixes τ_0, its minimal polynomial $30\tau^2 + 30\tau + 11$ implies

$$c = 30k, \qquad d - a = 30k, \qquad b = -11k$$

for some integer k. Its determinant is

$$(a + 15k)^2 + 105k^2.$$

For this to be a positive divisor of 30, $k = 0$, and the matrix is scalar. Thus the stabilizer is trivial as a group of fractional linear transformations. The quotient map is unramified at τ_0; since T is a degree-one coordinate on the quotient, $T'(\tau_0) \neq 0$. □

Proposition 7.2 (A logarithmic derivative identity)**.** *At $\tau = \tau_0$,*

$$\frac{D^2T}{DT} + \frac{61}{315}DT = \frac{15}{\pi\sqrt{105}} \qquad (\tau = \tau_0). \tag{7.6}$$

Proof. Let $A_0 = T'(\tau_0)$ and $B_0 = T''(\tau_0)$. Direct differentiation of the fractional linear transformation gives

$$\gamma'(\tau_0) = -1, \qquad \gamma''(\tau_0) = -\frac{60\mathrm{i}}{\sqrt{105}}.$$

For $X(\tau) = T(\tau)$ and $Y(\tau) = T(\gamma\tau)$, their first derivatives at τ_0 are $A_0, -A_0$, and their second derivatives are $B_0, B_0 + \gamma''A_0$. Differentiating $\Phi(X(\tau), Y(\tau)) = 0$ twice and using symmetry gives

$$\Phi_X\left(B_0 + \frac{\gamma''A_0}{2}\right) + (\Phi_{XX} - \Phi_{XY})A_0^2 = 0,$$

with all partial derivatives evaluated at $(-6, -6)$. Both A_0 and $\Phi_X(-6, -6)$ are nonzero by Lemma 7.1 and (6.4). Division by $2\pi\mathrm{i}A_0\Phi_X$ and the definition of D therefore give (7.6). □

Before evaluating w, we match the continuation branches. On $\tau = -1/2 + \mathrm{i}y$, $y \geq \sqrt{105}/30$, write $q = -\rho$. For $0 < \rho \leq 1173/10000 < 3/25$ the product for u_* gives

$$\rho\frac{\mathrm{d}}{\mathrm{d}\rho}\log|u_*| \leq -1 + 3\rho + \frac{3\rho^7(7 - 6\rho)}{(1 - \rho)^2} < 0.$$

For the last inequality one can replace each positive term by its bound at $\rho = 3/25$ and check the resulting rational inequality. As ρ increases from 0 to its value at τ_0, $|u_*|$ decreases from infinity to a value greater than one. Hence $T = 6 - |u_*| - |u_*|^{-1}$ increases from $-\infty$ to -6, and z moves from 0^- to $-1/6$ along the negative real axis. Since $P_0(z) > 0$ throughout this segment, the continued square root is the positive real one. Theorem 5.2 therefore continues along exactly the path of Corollary 4.2.

Corollary 7.3 (The required period evaluation)**.** *On the specified continuation branch,*

$$(7\theta_z + 5)w(-1/6) = \frac{6\sqrt{15}}{5\pi}. \tag{7.7}$$

Proof. Set $J(T) = 1/(T\sqrt{P_0(1/T)})$. The period formula reads $w = -JDT$, and $\theta_z = -T\,\mathrm{d}/\mathrm{d}T$. At $T = -6$,

$$P_0(-1/6) = \frac{175}{36}, \qquad J = -\frac{1}{5\sqrt{7}}, \qquad \frac{J'}{J} = \frac{47}{630},$$

where J' denotes differentiation with respect to T. The chain rule gives

$$\theta_z w = TJ'DT + TJ\frac{D^2T}{DT}.$$

Substitution of (7.6) yields

$$\begin{aligned}(7\theta_z + 5)w &= 7TJ\frac{15}{\pi\sqrt{105}} + \left(7TJ' - 5J - 7TJ\frac{61}{315}\right)DT \\ &= \frac{6\sqrt{15}}{5\pi}.\end{aligned}$$

Indeed, the coefficient of DT divided by J is $-42(47/630) - 5 + 42(61/315) = 0$. □

8 Proof of Sun's conjecture

Proof of Theorem 1.1. Differentiate (4.9); here $z = -r/2$, so θ_r acting on $w(-r/2)$ is θ_z. At $r = 1/3$,

$$\frac{\theta_r\sqrt{1+2r}}{\sqrt{1+2r}} = \frac{r}{1+2r} = \frac{1}{5}.$$

Consequently,

$$(35\theta_r + 18)g(1/3) = 5\sqrt{\frac{5}{3}}\,(7\theta_z + 5)w(-1/6) = \frac{30}{\pi},$$

where the last equality is Corollary 7.3. Corollary 3.2 now gives $S = 3(30/\pi) = 90/\pi$. Proposition 2.1 supplies the asserted absolute convergence and validates every rearrangement used to identify S. This proves the identity conjectured by Sun. □

A The complete modular polynomial

Write

$$\Phi(X, Y) = X^8 + Y^8 + \sum_{i,j=0}^{7} a_{ij}X^iY^j.$$

The row index i and column index j start at zero. To keep every integer legible, the coefficient matrix is displayed in two column blocks:

$$(a_{ij}) = (M_0\ M_1). \tag{A.1}$$

$$M_0 = \begin{pmatrix} 0 & 0 & 451584 & 1843968 \\ 0 & -18432 & 1897728 & -768768 \\ 451584 & 1897728 & 4643296 & -1340192 \\ 1843968 & -768768 & -1340192 & -481600 \\ 1844752 & -357504 & -571928 & 395752 \\ -75488 & -286608 & 145768 & 34468 \\ 3528 & -15288 & 38059 & -23863 \\ -56 & 588 & -161 & -665 \end{pmatrix}.$$

$$M_1 = \begin{pmatrix} 1844752 & -75488 & 3528 & -56 \\ -357504 & -286608 & -15288 & 588 \\ -571928 & 145768 & 38059 & -161 \\ 395752 & 34468 & -23863 & -665 \\ -19768 & -38192 & 5145 & 539 \\ -38192 & 8512 & 63 & -161 \\ 5145 & 63 & -182 & 21 \\ 539 & -161 & 21 & -1 \end{pmatrix}.$$

The assembled matrix is symmetric. Substitution and differentiation give

$$\Phi_{XX}(-6,-6) = -723889152000, \qquad \Phi_{XY}(-6,-6) = -807091200000,$$

together with (6.3) and (6.4). Section 6 explains why the finite principal-part reconstruction determines this actual modular polynomial.

B Exact finite verification

The mathematical certificates are the explicit rational identity (3.4), the normal-derivative recursion (4.4)–(4.5), the operator factorization (4.6), the gauge identity (4.10), and the principal-part algorithm with Newton recursion (6.7). All their input data appear above. Clearing denominators in rational identities, or comparing coefficients of the specified operators and polynomials, reduces each certificate to finitely many integer checks.

The directory `anc/` accompanying the source supplies executable checks in Python with SymPy. Its entry point is `anc/verify_certificates.py`. From that directory, run

```
python verify_certificates.py.
```

The nine checks cover the generic exact differential; the target-point coordinate correction; the normal-jet elimination and factorization; an alternative differentiated-Horn elimination; the period gauge; eta multipliers and cusp orders; the Hecke cosets and CM matrix; the modular-polynomial reconstruction; and the rational CM enclosure. The gauge check uses the explicit coefficients of (4.3), without a fitted-coefficient input file.

The modular-polynomial algorithm is particularly short to reproduce:

1. Truncate $f = \prod_{1\le n\le 64,\,(n,30)=1}(1-q^n)^3$ modulo q^{65}, compute its formal inverse, and form $qT = f + 6q + q^2/f$ to the same precision.

2. For each $1 \le m \le 8$, extract the nonpositive part in (6.6). Starting at degree $7m$, subtract multiples of powers of T to recover S_m.

3. Apply (6.7) and compare the resulting coefficients with Appendix A; then factor the diagonal and take the specified derivatives.

The degree bound in Section 6 makes this a proof by finite exact computation. The analytic and modular arguments in the main text are also necessary; numerical series matching alone would not replace them.

References

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孙智伟一个二项式级数猜想的证明

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摘要

本文证明孙智伟提出的一个双重二项式级数猜想，其精确值为 $90/\pi$。利用常数项积分和一个显式有理恰当微分恒等式，将所讨论的双重二项式级数化为一条特殊曲线上的周期及其导数。通过原系数满足的 Horn 方程作精确消元，将该周期识别为 Lian–Yau 的 level-30 模周期；再用一个可由有限主部严格确定的 Hecke–7 模方程，求出所需特殊值及导数。全过程包含绝对收敛、解析延拓、平方根分支和有限计算充分性的说明。

1 引言与记号

孙智伟在文献 [1, 印刷第 11 页] 的猜想 1.7 之后提出了下述级数恒等式。本文的目的是证明这一猜想。证明先将双变量级数化为限制周期，再利用 Lian–Yau 的第 30 级模周期恒等式及一个显式 Hecke 模方程求值。全文给出所需有限计算的完整数据及其充分性界。

定理 1.1 (孙智伟的二项式级数猜想)**.** 下式成立：

$$\begin{aligned}\sum_{n=0}^{\infty}\frac{357n+103}{2160^n}\binom{2n}{n}a_n&=\frac{90}{\pi},\\ a_n&=\sum_{k=0}^{n}\binom{n}{k}\binom{n+2k}{2k}\binom{2k}{k}(-324)^{n-k}.\end{aligned}\tag{1.1}$$

内层为有限求和，外层取通常实数意义下的部分和极限；事实上，将所有以 (n,k) 为指标的求和项取绝对值后仍收敛。

约定 m,n,k 为非负整数，$\binom{n}{k}=n!/(k!(n-k)!)$ $(0\le k\le n)$。记 $\theta_r=r\,\mathrm{d}/\mathrm{d}r$，$[x^j]$ 表示系数提取，$\mathrm{CT}_{x,y}$ 表示所指定 Laurent 展开中的常数项。对 F 的偏导数始终固定另一个原变量；微分算子的乘法表示复合，并包括对系数求导。记 $\mathbb{H}=\{\tau\in\mathbb{C}:\operatorname{Im}\tau>0\}$。正行列式矩阵按分式线性变换作用于 $\mathbb{H}$，相差非零标量的矩阵表示同一变换。所有平方根均按文中规定的解析分支取值。

2 双变量级数、常数项表示与绝对收敛

定义

$$F(u,v)=\sum_{m,k\ge0}\binom{2m+2k}{m+k}\frac{(m+3k)!}{m!(k!)^3}u^mv^k,\qquad \Theta=u\partial_u+v\partial_v.\tag{2.1}$$

由

$$\binom{n}{k}\binom{n+2k}{2k}\binom{2k}{k}=\frac{(n+2k)!}{(n-k)!(k!)^3}$$

以及代换 $m=n-k$，原式左端可写成

$$S=(357\Theta+103)F\left(-\frac{3}{20},\frac{1}{2160}\right). \tag{2.2}$$

上述重排及逐项求导的合法性将在下面一并证明。

命题 2.1 (常数项表示与绝对收敛)**.** 令 $a=1-x-y$，定义域

$$\mathcal{D}=\left\{(u,v)\in\mathbb{C}^2:4\left(|u|+\frac{10000}{99}|v|\right)<\frac{4}{5}\right\}.$$

在 $\mathcal{D}$ 上，F 的级数及其乘以任意固定多项式权重的级数均绝对且局部一致收敛，且有

$$F(u,v)=\mathrm{CT}_{x,y}\,\frac{1}{a}\left(1-\frac{4(u+v/(xy))}{a}\right)^{-1/2}. \tag{2.3}$$

其中内层平方根在 $u=v=0$ 处取值 1。因此 (2.2) 的重排和逐项求导合法。

证明. 令 $a=1-x-y$，先对 u,v 作二项展开。右端的 u^mv^k 系数为

$$\binom{2m+2k}{m+k}\binom{m+k}{k}[x^ky^k](1-x-y)^{-m-k-1},$$

而

$$[x^ky^k](1-x-y)^{-m-k-1}=\frac{(m+3k)!}{(m+k)!(k!)^2},$$

故它恰为 (2.1) 中的系数。

固定正向乘积环面

$$\mathcal{T}=\{(x,y):|x|=9/100,\ |y|=11/100\}.$$

在该环面上 $|x+y|\le 1/5$，$|xy|=99/10000$。在目标参数处有严格不等式

$$\frac{4(|u|+|v|/|xy|)}{1-|x|-|y|}=\frac{10519}{10692}<1. \tag{2.4}$$

因此 (2.3) 中的二项展开以及 $(1-x-y)$ 的负幂展开均绝对且一致收敛。将所有项换成绝对值后，同一正项控制级数还说明：(2.1) 的双重级数及其乘上任意固定的 $(m+k)$ 的多项式权重后，均绝对收敛。更明确地，对 $\mathcal{D}$ 中任一参数，稍微增大 $|u|,|v|$ 后仍留在 $\mathcal{D}$ 中，所增加的指数因子可控制任意固定多项式权重。故在紧子集上有一致控制。故 (2.2) 的重排及逐项求导均合法，而且

$$F(u,v)=\frac{1}{(2\pi\mathrm{i})^2}\int_{\mathcal{T}}\frac{\mathrm{d}x\wedge\mathrm{d}y}{xy\sqrt{(1-x-y)(1-x-y-4u-4v/(xy))}}. \tag{2.5}$$

这里的分母平方根严格指

$$(1-x-y)\left(1-\frac{4(u+v/(xy))}{1-x-y}\right)^{1/2},$$

其中使用二项级数分支；它在所用环面附近单值且不为零。 □

考虑参数曲线

$$U(r)=-\frac{r(2+3r)}{4(1+2r)},\qquad V(r)=\frac{r^4}{16(1+2r)},\qquad g(r)=F(U(r),V(r)). \tag{2.6}$$

当 $0\le r\le 1/3$ 时，$|U(r)|$ 和 $V(r)$ 均单调增加，终点分别为 $3/20$ 和 $1/2160$。所以同一环面及 (2.4) 证明 g 及所需偏导数在整个区间 $[0,1/3]$ 的一个复邻域内解析。

3 原径向导数与曲线切向导数的关系

引理 3.1. 沿曲线 (2.6) 有

$$Cg+\alpha F_u(U,V)+\beta F_v(U,V)=0, \tag{3.1}$$

其中

$$\begin{aligned} C&=3r(1+2r)(r+2),\\ \alpha&=-\frac{3r(3r+2)(r^2+6r+2)}{4},\\ \beta&=\frac{3r^4(3r^2+4r-2)}{16}. \end{aligned} \tag{3.2}$$

该恒等式在零点附近以及整个区间 $0\le r\le 1/3$ 上均成立。

证明. 先设 $0<r\le 1/3$。在 (2.5) 中记

$$s=x+y,\quad p=xy,\quad c=1-4U,\quad b=4V,\quad \mathcal{W}=p\sqrt{(1-s)(c-s-b/p)},$$

并令

$$X=(c-s)p,\qquad Y=(c-s)(x-y)\mathcal{W}.$$

直接计算可得

$$\begin{aligned} Y^2=P(s,X)&:=(1-s)X\{s^2(c-s)-4X\}(X-b),\\ \frac{\mathrm{d}s\wedge\mathrm{d}X}{Y}&=\frac{\mathrm{d}x\wedge\mathrm{d}y}{\mathcal{W}}. \end{aligned} \tag{3.3}$$

后一个等式是微分形式的拉回恒等式。积分始终在原来的环面上进行，因而无须另乘代数映射的次数。

现给出所需的显式有理证书。定义

$$\begin{aligned} q_s&=-\frac{r^2}{4(r+1)^2}, & \sigma&=-\frac{r^2}{1+2r},\\ X_0&=q_s s(s-c)(s-1), & h&=s(s-c)(s-1)(s-\sigma)(s-r-1),\\ K&=\frac{3r^2(2r+1)^2}{2(r+1)^3}, & R_0&=-\frac{3(2r+1)(s-1)(r^3+r^2+4rs+2s)}{r+1}, \end{aligned}$$

以及

$$\begin{aligned} T_1&=-\frac{3(2r+1)(3r^3+7r^2+12rs-4r+6s-4)}{r+1},\\ T_0&=\frac{3r^2(r-s+1)^2(r^2+2rs+s)(6rs-3r+3s-1)}{2(r+1)^3},\\ R&=\frac{Kh}{X-X_0}+R_0,\\ \mathcal{A}&=\frac{K(\partial_s X_0)h}{X-X_0}+T_1X+T_0-\frac{4\alpha X}{c-s}. \end{aligned}$$

以 r,s,X 为独立变量，有如下有理函数恒等式：

$$R_s+\mathcal{A}_X-\frac{RP_s+\mathcal{A}P_X}{2P}=C+\frac{2\alpha X}{(c-s)(X-b)}+\frac{2\beta}{X-b}. \tag{3.4}$$

这是清除分母后可以逐项验证的有限多项式恒等式。附录B 列出了精确验证程序。特别要注意，$\mathcal{A}$ 中的 $-4\alpha X/(c-s)$ 不能省略：原变量不变时 $X_u=-4X/(c-s)$。不加此项得到的是固定 X 的导数，并不是原积分中的偏导数。其校正也可由下式直接核验：

$$\frac{2s^2}{s^2(c-s)-4X}-\frac{4}{c-s}+\frac{2XP_X}{(c-s)P}=\frac{2X}{(c-s)(X-b)}. \tag{3.5}$$

令一形式

$$\Omega=\frac{R\,\mathrm{d}X-\mathcal{A}\,\mathrm{d}s}{Y}.$$

由 (3.4)，其外微分为该式右端乘以 $\mathrm{d}s\wedge\mathrm{d}X/Y$。该一形式在所选环面附近单值且无极点。事实上，(2.4) 保证 $\mathcal{W}\neq0$；两条圆周半径不同保证 $x-y\neq0$；此外 $1-s,c-s,X-b$ 均不为零。对于唯一新出现的分母，有 $|q_s|\le1/64$，从而

$$\left|\frac{X_0}{c-s}\right|=|q_s s(s-1)|\le\frac{3}{800}<\frac{99}{10000}=|p|.$$

故 $X-X_0\neq0$。这些都是严格不等式，因而在相应复邻域中仍成立。由闭环面上的 Stokes 定理，$\int_{\mathcal{T}}\mathrm{d}\Omega=0$。

另一方面，在原来的 x,y 坐标下求导，F_u 和 F_v 相对于 (3.3) 中二形式的乘子分别为

$$\frac{2X}{(c-s)(X-b)},\qquad\frac{2}{X-b}.$$

因此对 (3.4) 积分便得到 (3.1)。由于固定环面积分、所需偏导数及 C,α,β 在零点附近解析，恒等式也由解析性延拓至 $r=0$。 □

推论 3.2 (目标点的导数转换)**.** 在 $r=1/3$、$(u,v)=(U(1/3),V(1/3))$ 处，

$$\Theta F=\frac{15\theta_r g-7g}{51},\qquad S=3(35\theta_r+18)g(1/3).$$

证明. 在 $r=1/3$ 处，(3.1) 化为

$$\frac{35}{9}g-\frac{37}{12}F_u-\frac{1}{1296}F_v=0.$$

由 (2.6) 直接求导又有

$$\theta_r g=-\frac{7}{50}F_u+\frac{1}{600}F_v,\qquad\Theta F=-\frac{3}{20}F_u+\frac{1}{2160}F_v.$$

消去两个偏导数，得到

$$\Theta F=\frac{15\theta_r g-7g}{51},\qquad S=3(35\theta_r+18)g(1/3). \tag{3.6}$$

□

4 限制周期的微分方程

记 $A = u\partial_u$、$B = v\partial_v$。由 (2.1) 中相邻系数的阶乘比值可直接得到 Horn 方程

$$\begin{aligned} &H_1 F = 0, \quad H_1 = A(A+B) - 2u(2A+2B+1)(A+3B+1), \\ &H_2 F = 0, \quad H_2 = B^3(A+B) \\ &\qquad\qquad - 2v(2A+2B+1)\prod_{j=1}^{3}(A+3B+j). \end{aligned} \tag{4.1}$$

坐标边界上的系数也由左端的 Euler 因子消去，故方程适用于完整 Taylor 级数。

命题 4.1 (限制后的微分方程)**.** *函数 g 满足 $L_3 g = 0$，其中*

$$L_3 = \sum_{j=0}^{3} p_j(r)\theta_r^j, \tag{4.2}$$

而

$$\begin{aligned} p_0 &= \frac{r(144r^5 + 159r^4 + 9r^3 - 45r^2 - 16r - 2)}{48}, \\ p_1 &= \frac{r(2r+1)(312r^4 + 151r^3 - 150r^2 - 106r - 18)}{96}, \\ p_2 &= \frac{r(2r+1)^3(18r^2 - 13r - 10)}{32}, \\ p_3 &= \frac{(r-2)(2r+1)^3(3r+2)(4r+1)}{96}. \end{aligned} \tag{4.3}$$

证明. 下面给出精确消元的定义和证书，避免将有限个展开系数的吻合当作证明。引入法向坐标

$$u = U(r), \qquad v = V(r) + t, \qquad a = \frac{1}{U'}, \quad d = -\frac{V'}{U'}.$$

由于

$$U' = -\frac{3r^2 + 3r + 1}{2(2r+1)^2} \neq 0,$$

坐标变换在所用区间附近有效，且 $\partial_u = a\partial_r + d\partial_t$、$\partial_v = \partial_t$。将 H_1 写成普通偏导数的线性组合，其非零系数为

$$\begin{aligned} &c_{20} = (1-4u)u^2, \quad c_{11} = (1-16u)uv, \quad c_{02} = -12uv^2, \\ &c_{10} = u(1-10u), \quad c_{01} = -22uv, \qquad\quad c_{00} = -2u. \end{aligned}$$

变换到 r, t 坐标后，六个系数是

$$\begin{aligned} C_{20} &= c_{20}a^2, & C_{11} &= 2c_{20}ad + c_{11}a, \\ C_{02} &= c_{20}d^2 + c_{11}d + c_{02}, & C_{10} &= c_{20}aa' + c_{10}a, \\ C_{01} &= c_{20}ad' + c_{10}d + c_{01}, & C_{00} &= c_{00}. \end{aligned}$$

具体地，

$$C_{02}(r,0) = \frac{3r^8(r+1)(3r+2)(9r^2+8r+2)}{256(2r+1)^2(3r^2+3r+1)^2} \neq 0 \quad (0 < r \leq 1/3).$$

对 $(i,j)\neq(0,2)$ 定义 $Q_{ij}=-C_{ij}/C_{02}$，并定义作用于 g 的普通微分算子 J_j：

$$J_0 g = g, \qquad J_1 g = -\frac{Cg+\alpha a g'}{\alpha d+\beta}, \tag{4.4}$$

$$J_{n+2} g = \sum_{(i,j)\neq(0,2)} \sum_{k=0}^{n} \binom{n}{k} (\partial_t^k Q_{ij})(r,0) \left(\frac{\mathrm{d}}{\mathrm{d}r}\right)^i J_{n-k+j} g, \qquad n=0,1,2. \tag{4.5}$$

这里求和中的 (i,j) 取上述六个系数的指标，并排除 $(0,2)$。先求导再取 $t=0$。式 (4.4) 来自引理3.1，而 (4.5) 来自把 $H_1 F=0$ 解出二阶法向导数后，再求 n 次法向导数。故 $J_j g$ 恰是 $F(U(r),V(r)+t)$ 在 $t=0$ 处的第 j 阶法向导数，$j\le 4$。所用分母确为非零有理函数；例如

$$\alpha d+\beta = -\frac{15 r^4 (r+1)(9r^2+8r+2)}{16(3r^2+3r+1)}.$$

现将因子化的 H_2 变换到 r,t 坐标，再用 J_j 替换所有法向导数。记所得普通微分算子为 N，则 $Ng=0$。令 $D_0=9r^2+8r+2$，有限有理运算给出如下精确算子恒等式：

$$N = \left(q_1 \frac{\mathrm{d}}{\mathrm{d}r} + q_0\right) L_3, \tag{4.6}$$

其中

$$\begin{aligned} q_1 &= -\frac{3r(2r+1)}{5D_0^2}, \\ q_0 &= \frac{r(1944r^5+5769r^4+6588r^3+3675r^2+1010r+110)}{15(r+1)(2r+1)^2 D_0^3}. \end{aligned} \tag{4.7}$$

算子系数均置于导数左侧，(4.6) 的乘法为算子复合，包含系数的求导。上述递推及 (4.6) 给出了全部有限验证所需的数据；`anc/research/independent_ck.py` 精确核验全部五个普通导数系数；另一个微分关系消元证书亦在附件中提供。

还须排除 (4.6) 多出的第一阶因子。令 $E=L_3 g$。在零点的去心邻域内，$E'=-(q_0/q_1)E$。更明确地，

$$-\frac{q_0}{q_1} = \frac{1944r^5+5769r^4+6588r^3+3675r^2+1010r+110}{9(r+1)(2r+1)^3(9r^2+8r+2)}$$

在零点解析。由于 g 解析以及 $L_3|_{r=0}=-\theta_r^3/24$，有 $E(0)=0$。因而一阶方程延拓到零点，由解析初值问题的唯一性得 $E\equiv 0$。这便在零点附近证明了 $L_3 g=0$；再由 $L_3 g$ 在整个区间的复邻域内解析，恒等定理将其延拓至 $[0,1/3]$。 □

定义 $w(z)$ 为下列三阶方程在零点满足 $w(0)=1$ 的 Taylor 解，记 $\theta_z = z\,\mathrm{d}/\mathrm{d}z$：

$$\begin{aligned} \mathcal{L} = {} & \theta_z^3 - z(1+2\theta_z)(4+7\theta_z+7\theta_z^2) \\ & + z^2(1+\theta_z)(72+106\theta_z+53\theta_z^2) \\ & - 14 z^3 (1+\theta_z)(2+\theta_z)(3+2\theta_z) \\ & - 24 z^4 (2+\theta_z)(3+2\theta_z)(5+2\theta_z), \qquad \mathcal{L} w = 0. \end{aligned} \tag{4.8}$$

推论 4.2 (周期识别)**.** *沿 $0\le r\le 1/3$，有*

$$g(r) = \sqrt{1+2r}\, w(-r/2),$$

其中平方根为正实分支，w 按负实轴作解析延拓。

证明. 方程 (4.8) 对第 $n \geq 1$ 阶新 Taylor 系数的乘子为 n^3，故 $w(0) = 1$ 唯一确定 Taylor 解；正则奇点的 Frobenius 理论保证局部收敛。把 $z = -r/2$ 代入并逐项求导，可直接核验

$$L_3\{\sqrt{1+2r}\, w(-r/2)\} = -\frac{(1+2r)^{5/2}}{24}(\mathcal{L}w)(-r/2). \tag{4.9}$$

在 L_3 的解析解递推中，新系数在第 n 阶的乘子为 $-n^3/24$，$n \geq 1$。故常数项唯一确定解析解。比较零点的值即得

$$g(r) = \sqrt{1+2r}\, w(-r/2). \tag{4.10}$$

普通最高阶导数的系数为 $r^3 p_3(r)$，它在 $0 < r \leq 1/3$ 上不为零。因此 (4.10) 可沿整个区间延拓；此处 $w(-1/6)$ 表示该延拓值，并非要求零点处的 Taylor 级数在 $-1/6$ 直接收敛。

□

5 模参数化及其归一化

令 $\tau \in \mathbb{H}$、$q = e^{2\pi \mathrm{i}\tau}$，并定义 Dedekind eta 函数

$$\eta(\tau) = e^{\pi \mathrm{i}\tau/12} \prod_{n \geq 1} (1 - q^n).$$

记

$$\begin{gathered} u_*(\tau) = \left[\frac{\eta(\tau)\eta(6\tau)\eta(10\tau)\eta(15\tau)}{\eta(2\tau)\eta(3\tau)\eta(5\tau)\eta(30\tau)}\right]^3, \\ T = u_* + 6 + u_*^{-1}, \qquad z = T^{-1}, \qquad D = \frac{1}{2\pi \mathrm{i}} \frac{\mathrm{d}}{\mathrm{d}\tau} = q \frac{\mathrm{d}}{\mathrm{d}q}, \\ P_0(z) = (1 - 8z)(1 - 4z)(1 + z)(1 - 3z). \end{gathered} \tag{5.1}$$

定理 5.1 (Lian–Yau 模周期恒等式)**.** 设 $\operatorname{Im}\tau$ 充分大。对以上显式定义的 z 及归一化 $w(0) = 1$，有

$$w(z(\tau)) = \frac{Dz}{z\sqrt{P_0(z)}}. \tag{5.2}$$

其中 $\sqrt{P_0(z)}$ 在 $z = 0$ 附近趋于 1。

证明. 准确出处为 Lian–Yau [2] 印刷页第 9 页的 $\Gamma_0(30)^+$ 表项，以及式 (4.6)、(4.7) 和定理 4.2 的式 (4.14)。该表项的算子正是 (4.8)。这里 $\Gamma_0(30)^+$ 指加入全部 Atkin–Lehner 对合所生成的群，平方根在尖点附近取值趋于 1 的分支。

下面核对显式函数 T 确实具有该文所需的群和归一化。对于无平方因子的 30，标准 Atkin–Lehner 陪集为

$$\Gamma = \bigcup_{e \mid 30} \Gamma_0(30) W_e,$$

其中每个陪集有行列式为 e、左下元素被 30 整除的整数矩阵代表。在尖点分母上，W_e 的作用为 $d \mapsto ed/\gcd(e,d)^2$；因此八个尖点构成一个轨道。

对于 $\delta = 1, 2, 3, 5, 6, 10, 15, 30$，$u_*$ 的 eta 指数 r_δ 依次为

$$(3, -3, -3, -3, 3, 3, 3, -3).$$

它们满足

$$\sum r_\delta = 0, \quad \sum \delta r_\delta = -24, \quad \sum (30/\delta) r_\delta = 24, \quad \prod \delta^{r_\delta} = 1.$$

eta 商判据及尖点阶数公式 [3, 定理 1.64–1.65]，亦见 [4, 定理 1.1–1.2]，说明 u_* 是 $\Gamma_0(30)$ 上平凡特征的权零模函数。这里仅用其变换性质及尖点亚纯性，允许负 eta 指数。尖点分母 $d \mid 30$ 处的阶数为

$$\frac{30}{24d} \sum_{\delta \mid 30} \frac{\gcd(d,\delta)^2}{\delta} r_\delta .$$

按上述八个分母排序，所得阶数依次是 $(1,-1,-1,-1,1,1,1,-1)$。

取矩阵

$$W_2 = \begin{pmatrix} 8 & 1 \\ 30 & 4 \end{pmatrix}, \quad W_3 = \begin{pmatrix} 3 & 2 \\ 30 & 21 \end{pmatrix}, \quad W_5 = \begin{pmatrix} 25 & 4 \\ 30 & 5 \end{pmatrix}. \tag{5.3}$$

Dedekind 的 eta 变换公式 [5, §23.18] 给出

$$u_*(W_e \tau) = u_*(\tau)^{-1}, \qquad e = 2, 3, 5. \tag{5.4}$$

为使乘子问题可精确复核，写 $W_e = (ea, b; 30c, ed)$。对每个 $\delta \mid 30$，令 $h_\delta = \gcd(\delta, e)$、$m = \delta e / h_\delta^2$，则

$$\delta W_e \tau = \begin{pmatrix} h_\delta a & (\delta/h_\delta) b \\ 30 h_\delta c/(e\delta) & ed/h_\delta \end{pmatrix} (m\tau),$$

其中矩阵属于 $\mathrm{SL}_2(\mathbb{Z})$。将 eta 变换公式中的 Dedekind 和按 r_δ 加权，对三个 e 所得乘子指数都恰为零，且 $\prod h_\delta^{r_\delta} = 1$；置换 $\delta \mapsto m$ 则将全部 eta 指数取负。这就是 (5.4) 的有限有理验证。

于是 T 对 $\Gamma = \Gamma_0(30)^+$ 不变。由于 30 无平方因子，八个尖点在全部 Atkin–Lehner 对合下属于同一轨道。eta 在上半平面无零点，故 T 在内部无极点。在每个原尖点处，u_* 和 u_*^{-1} 中恰有一个具有一阶极点，故不存在消去。下降后的唯一尖点也是一阶极点：否则拉回后的极点阶数不可能为一。因而 T 从紧化商曲面到 $\mathbb{P}^1$ 的次数为一，证明它是该群的 Hauptmodul（即模函数域的生成元）。特别地，只有尖点极点的模函数都是 T 的多项式。

由 eta 乘积还可得

$$u_* = q^{-1} \prod_{\gcd(n,30)=1} (1-q^n)^3, \qquad T = q^{-1} + 3 + 4q + 2q^2 + O(q^3). \tag{5.5}$$

必须先固定镜像映射的首项。Lian–Yau 文献的式 (4.12)–(4.13) 采用 Frobenius 归一化

$$q = z \exp\!\left(\frac{h_0(z)}{\widetilde{w}(z)} \right), \qquad h_0(0) = 0, \quad \widetilde{w}(0) = 1,$$

故 $z = q + O(q^2)$；这里 h_0 与前文有理证书中的 h 无关。这排除了任意常数倍的 q 归一化。随后 (4.8) 给出 $w(z) = 1 + 4z + O(z^2)$，结合 (5.2) 及 $P_0(z) = 1 - 14z + O(z^2)$，得到 $z = q - 3q^2 + O(q^3)$。两个在唯一尖点有相同一阶极点、相同首项及常数项的 Hauptmodul 必相等。这就核对了 (5.2) 所用的正是 (5.1) 中的 T。

□

6 由有限主部严格确定的 Hecke–7 模方程

使用八个行列式为七的矩阵

$$H_0=\begin{pmatrix}7&0\\0&1\end{pmatrix},\qquad H_{j+1}=\begin{pmatrix}1&j\\0&7\end{pmatrix},\quad 0\le j\le 6. \tag{6.1}$$

命题 6.1 (Hecke 模多项式)**.** 下式定义唯一的 $\Phi(X,Y)\in\mathbb{Z}[X,Y]$：

$$\Phi(T(\tau),Y)=\prod_{i=0}^{7}\{Y-T(H_i\tau)\}.$$

它关于每个变量的次数均为八，关于 X,Y 对称，其完整系数见附录A；对角分解与指定偏导数分别由 (6.6)、(6.7) 给出。

证明. 因为 7 与 30 互素，通常的 Hecke 陪集分解说明 $\Gamma_0(30)$ 置换这八个函数 $T(H_i\tau)$。同样的结论对 W_2,W_3,W_5 也成立。具体地，按 $i=0,\dots,7$ 排列，三个置换依次是

$$\begin{aligned}\sigma_2&=(3,2,5,7,0,6,1,4),\\ \sigma_3&=(1,4,7,0,2,5,6,3),\\ \sigma_5&=(7,2,6,1,5,3,0,4).\end{aligned} \tag{6.2}$$

这里列表表示 $\sigma_e(i)$ 的八个值，而非循环记号。逐一计算的二十四个矩阵 $H_iW_eH_{\sigma_e(i)}^{-1}W_e^{-1}$ 均为行列式一、左下角被三十整除的整矩阵。

因此定义式

$$\Phi(T(\tau),Y)=\prod_{i=0}^{7}\{Y-T(H_i\tau)\} \tag{6.3}$$

的系数是 Γ 上内部无极点的模函数，因而是 T 的多项式。在唯一尖点处，八个根的 Puiseux 极点阶数为 7 和七个 $1/7$。第 k 个初等对称系数的极点阶数至多为 $7+(k-1)/7$；由于该阶数必须为整数，其多项式次数在 $1\le k\le 7$ 时至多为七，在 $k=8$ 时至多为八。

为完整说明有限计算的充分性，令 S_m 是八个根的第 m 个幂和。它是 T 的次数不超过 $7m$ 的多项式。其主部（包括常数项）等于下式的非正次幂部分：

$$T(q^7)^m+7\sum_{n}[q^{7n}]T(q)^m\,q^n. \tag{6.4}$$

第二项来自七个单位根的筛选。一个 $T=q^{-1}+3+\cdots$ 的次数不超过 $7m$ 的多项式，由其非正 Laurent 系数唯一确定。因此从最高负次幂开始逐项减去 T 的幂，便能精确重建 S_m。对 $m=1,\dots,8$ 作此运算，再用 Newton 恒等式得到全部初等对称系数，便得到真实的模方程 Φ。

Newton 递推明确写为

$$E_0=1,\qquad kE_k=\sum_{j=1}^{k}(-1)^{j-1}E_{k-j}S_j,\qquad \Phi=\sum_{k=0}^{8}(-1)^kE_k(X)Y^{8-k}. \tag{6.5}$$

所需的最大次数为 56。计算 $T^d=q^{-d}(qT)^d$ 的非正次幂只需 qT 到第 d 阶，故把 qT 算到 q^{64} 已充分；所有乘法、消元和 Newton 除法均在整数中精确进行。附录A 给出了完整系数矩阵，附录B 给出了重建程序。

所得多项式满足 $\Phi(X,Y)=\Phi(Y,X)$，并且

$$\begin{aligned}\Phi(X,X)=\ &-X^2(X-24)(X-6)(X-4)^2(X+2)^2(X+6)\\ &\cdot(X^2-4X-4)(X^3-10X^2+28X+4),\end{aligned}\tag{6.6}$$

以及

$$\Phi_X(-6,-6)=429649920000,\qquad \frac{\Phi_{XX}-\Phi_{XY}}{\Phi_X}(-6,-6)=\frac{61}{315}.\tag{6.7}$$

这些结论都是附录中有限多项式的精确恒等式。这里 “Hecke–7” 指同源映射的次数，Φ 对每个变量的次数则为八。

□

7 特殊点的精确取值及导数

令

$$\tau_0=\frac{-15+\mathrm{i}\sqrt{105}}{30},\qquad \gamma=\begin{pmatrix}-15&-11\\30&15\end{pmatrix}.\tag{7.1}$$

矩阵 γ 的行列式为 105，且 $\gamma\tau_0=\tau_0$。又有精确分解

$$\begin{aligned}\gamma&=W_{15}M_7,\qquad W_{15}=\begin{pmatrix}15&7\\30&15\end{pmatrix},\\ M_7&=\begin{pmatrix}-29&-18\\60&37\end{pmatrix}=\begin{pmatrix}-29&14\\60&-29\end{pmatrix}H_5.\end{aligned}\tag{7.2}$$

其中 W_{15} 是 Atkin–Lehner 矩阵，最后一个行列式一的矩阵属于 $\Gamma_0(30)$。故

$$\Phi(T(\tau),T(\gamma\tau))=0.\tag{7.3}$$

特别地，$T(\tau_0)$ 是 (6.6) 的某个根。

下面严格确定是哪个根。记

$$q(\tau_0)=-\rho,\qquad \rho=e^{-\pi\sqrt{105}/15}.$$

使用 $3.14<\pi<22/7$、$10.24<\sqrt{105}<10.25$，并分别用 e^{-x} 的三十一阶和三十二阶 Taylor 多项式给出下、上界，得到

$$\frac{1167}{10000}<\rho<\frac{1173}{10000}.\tag{7.4}$$

例如可取 $a=(314/100)(1024/100)/15$、$b=(22/7)(1025/100)/15$，直接用有理数验证

$$\frac{1167}{10000}<\sum_{j=0}^{31}\frac{(-b)^j}{j!},\qquad \sum_{j=0}^{32}\frac{(-a)^j}{j!}<\frac{1173}{10000}.$$

Taylor 余项的符号分别给出所需下、上界。

与三十互素的正整数均为奇数，故 (5.5) 给出

$$|u_*|=\rho^{-1}(1+\rho)^3\prod_{n\ge7,\,(n,30)=1}(1+\rho^n)^3.$$

令 $\rho_{\mathrm{hi}}=1173/10000$、$\rho_{\mathrm{lo}}=1167/10000$，$E_0=3\rho_{\mathrm{hi}}^{7}/(1-\rho_{\mathrm{hi}})$。尾积在 1 与 $(1-E_0)^{-1}$ 之间，因为 $\prod(1+a_j)\le e^{\sum a_j}\le(1-\sum a_j)^{-1}$。函数 $f(t)=(1+t)^3/t$ 在本区间递减，所以

$$f(\rho_{\mathrm{hi}})<|u_*|<\frac{f(\rho_{\mathrm{lo}})}{1-E_0}.$$

将这些有理界代入 $T=6-|u_*|-|u_*|^{-1}$，得到

$$-\frac{61}{10}<T(\tau_0)<-\frac{59}{10}. \tag{7.5}$$

这是有限有理不等式，精确验证见 `anc/research/cm_enclosure.py`。在此区间中，(6.6) 除 $X+6$ 外的因子均不为零：所列线性根均在区间外，二次因子的根 $2\pm2\sqrt{2}$ 均大于 -1，三次因子则因 $X<-5$ 而严格为负。因此

$$T(\tau_0)=-6,\qquad z(\tau_0)=-1/6. \tag{7.6}$$

在对模方程求导之前，须证明 $T'(\tau_0)\neq0$。Γ 中每个元素都有行列式为 $e\mid30$ 的整矩阵代表，且左下角被三十整除。设整矩阵 $\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)$ 固定 τ_0。由于其最小多项式为 $30\tau^2+30\tau+11$，有

$$c=30k,\qquad d-a=30k,\qquad b=-11k$$

对某个整数 k 成立。其行列式为

$$(a+15k)^2+105k^2.$$

若行列式是三十的正因子，必有 $k=0$，且矩阵为标量矩阵。所以该点无非平凡稳定子；而 T 是商曲面上的次数一坐标，因而它在该点不分歧，确有 $T'(\tau_0)\neq0$。

记 $A_0=T'(\tau_0)$、$B_0=T''(\tau_0)$。直接计算得到

$$\gamma'(\tau_0)=-1,\qquad \gamma''(\tau_0)=-\frac{60\mathrm{i}}{\sqrt{105}}.$$

对 (7.3) 求两次导数，利用 Φ 的对称性，得到

$$\Phi_X\left(B_0+\frac{\gamma''A_0}{2}\right)+(\Phi_{XX}-\Phi_{XY})A_0^2=0,$$

其中所有 Φ 的偏导数都在 $(-6,-6)$ 处取值。除以已经证明非零的 $2\pi\mathrm{i}A_0\Phi_X$，由 (6.7) 得

$$\frac{D^2T}{DT}+\frac{61}{315}DT=\frac{15}{\pi\sqrt{105}}\qquad(\tau=\tau_0). \tag{7.7}$$

还须核对平方根分支。在 $\tau=-1/2+\mathrm{i}y$、$y\ge\sqrt{105}/30$ 上，$q=-\rho$，且对 $0<\rho\le1173/10000<3/25$ 有

$$\rho\frac{\mathrm{d}}{\mathrm{d}\rho}\log|u_*|\le-1+3\rho+\frac{3\rho^7(7-6\rho)}{(1-\rho)^2}<0.$$

最后一个不等式甚至可将各正项用 $\rho\le3/25$ 放大后以有理数直接核验。故 $|u_*|$ 从无穷大递减到终点仍大于一的值，T 从 $-\infty$ 增至 -6，z 则沿负实轴从 0^- 到达 $-1/6$。在该区间上 $P_0(z)>0$，所以 (5.2) 的平方根就是正实平方根，与 (4.10) 的解析延拓分支一致。

令

$$J(T)=\frac{1}{T\sqrt{P_0(1/T)}}.$$

则 (5.2) 可写为 $w=-JDT$，且 $\theta_z=-T\,\mathrm{d}/\mathrm{d}T$；以下 J' 专指对 T 求导。在 $T=-6$ 处直接计算得

$$P_0(-1/6)=\frac{175}{36},\qquad J=-\frac{1}{5\sqrt{7}},\qquad \frac{J'}{J}=\frac{47}{630}.$$

链式法则给出

$$\theta_z w=TJ'DT+TJ\frac{D^2T}{DT}.$$

于是由 (7.7)，

$$\begin{aligned}(7\theta_z+5)w(-1/6)&=7TJ\frac{15}{\pi\sqrt{105}}+\left(7TJ'-5J-7TJ\frac{61}{315}\right)DT\\&=\frac{6\sqrt{15}}{5\pi}.\end{aligned}\tag{7.8}$$

这里 DT 的系数除以 J 恰为 $-42(47/630)-5+42(61/315)=0$。这一特殊值来自精确模方程及其导数，不依赖数值常数识别。

8 孙智伟猜想的证明

定理1.1的证明. 由 $z=-r/2$，作用于 $w(-r/2)$ 的 θ_r 就是 θ_z。并且 $\theta_r\sqrt{1+2r}/\sqrt{1+2r}=r/(1+2r)$，在 $r=1/3$ 时为 $1/5$。所以由 (4.10) 有

$$(35\theta_r+18)g=5\sqrt{\frac{5}{3}}\,(7\theta_z+5)w(-1/6)=\frac{30}{\pi},$$

其中最后一步使用 (7.8)。再代入 (3.6)，即得

$$S=3\cdot\frac{30}{\pi}=\frac{90}{\pi}.$$

这就证明了孙智伟提出的该级数猜想；命题2.1还保证绝对收敛及所用重排的合法性。 □

A 模多项式的完整系数

为使 (6.6) 和 (6.7) 可以独立复核，将完整多项式写成

$$\Phi(X,Y)=X^8+Y^8+\sum_{i,j=0}^{7}a_{ij}X^iY^j.$$

其中行指标为 i、列指标为 j，均从零开始。将矩阵按前四列与后四列分块，以保证每个整数清晰可读：

$$(a_{ij})=(M_0\ M_1).\tag{A.1}$$

$$M_0=\begin{pmatrix}
0 & 0 & 451584 & 1843968\\
0 & -18432 & 1897728 & -768768\\
451584 & 1897728 & 4643296 & -1340192\\
1843968 & -768768 & -1340192 & -481600\\
1844752 & -357504 & -571928 & 395752\\
-75488 & -286608 & 145768 & 34468\\
3528 & -15288 & 38059 & -23863\\
-56 & 588 & -161 & -665
\end{pmatrix}.$$

$$M_1 = \begin{pmatrix} 1844752 & -75488 & 3528 & -56 \\ -357504 & -286608 & -15288 & 588 \\ -571928 & 145768 & 38059 & -161 \\ 395752 & 34468 & -23863 & -665 \\ -19768 & -38192 & 5145 & 539 \\ -38192 & 8512 & 63 & -161 \\ 5145 & 63 & -182 & 21 \\ 539 & -161 & 21 & -1 \end{pmatrix}.$$

矩阵对称，故 Φ 对称。将 $Y = X$ 代入并分解，得到 (6.6)；逐项求导还给出

$$\Phi_{XX}(-6,-6) = -723889152000, \qquad \Phi_{XY}(-6,-6) = -807091200000,$$

与 (6.7) 一致。此矩阵由第6节说明的主部有限重建得到，故它的作用不限于给出一个在特殊点吻合的形式多项式。

B 精确有限验证

正文的计算证书包括有理恒等式 (3.4)、法向递推 (4.4)–(4.5)、算子分解 (4.6)、变换恒等式 (4.9) 以及主部算法与 Newton 递推 (6.5)。所有输入均已在正文给出；清除分母并逐项比较系数，就得到有限个整数恒等式。

源码附件中的 `anc/` 目录提供 Python/SymPy 精确核验程序，入口为 `anc/verify_certificates.py`。在该目录下运行

```
python verify_certificates.py
```

即可重现九项检查：通用恰当微分、目标点坐标校正、法向消元及算子分解、另一组微分 Horn 消元、周期变换、eta 乘子及尖点阶、Hecke 陪集及特殊点矩阵、模多项式重建、特殊点的有理包围。程序直接使用正文显式系数，不依赖历史拟合数据或审查记录。

模多项式重建可按以下步骤独立复现：

1. 将 $f = \prod_{1\le n\le 64,\,(n,30)=1}(1-q^n)^3$ 截断至模 q^{65}，求形式逆，并构造 $qT = f + 6q + q^2/f$。
2. 对 $m = 1,\dots,8$，提取 (6.4) 的非正次幂部分，从 $7m$ 次开始逐项减去 T 的幂，重建幂和 S_m。
3. 按 (6.5) 求全部 E_k，与附录A的系数比较，再进行对角因式分解和所需偏导计算。

第6节的次数界保证这一有限计算的充分性。正文关于解析性、模函数群、归一化及分支的论证同样是证明的必要部分，不能仅用有限阶展开的数值吻合替代。

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